

Practical 1

1.Environment Setup and JavaScript Basics

a. Install Node.js and Visual Studio Code, and verify installation.

```
PS C:\Users\itlab\Desktop\radha tyit\mern stact> node -v
v24.17.0
PS C:\Users\itlab\Desktop\radha tyit\mern stact> npm -v
11.13.0
PS C:\Users\itlab\Desktop\radha tyit\mern stact>
```

b. Create a simple Node.js file and run it using the terminal.

```
JS pract1b.js x
JS pract1b.js
1 console.log("Hello, welcome to 1st practical of MERN ");
2 console.log("Radhika");

PS C:\Users\itlab\Desktop\radha tyit\mern stact> node pract1b.js
Hello, welcome to 1st practical of MERN
Radhika
PS C:\Users\itlab\Desktop\radha tyit\mern stact>
```

c. Write a Node.js script demonstrating basic JavaScript commands (console log, variables, and arithmetic operations).

```
JS practic.js > ...
1 let num1 = 10;
2 let num2 = 2;
3 console.log("addition:", num1 + num2);
4 console.log("subtraction:", num1 - num2);
5 console.log("multiplication:", num1 * num2);
6 console.log("division:", num1 / num2);
7 console.log("radhika");

PS C:\Users\itlab\Desktop\radha tyit\mern stact> node practic.js
addition: 12
subtraction: 8
multiplication: 20
division: 5
radhika
```

d. Create and run a simple program using variables and functions.

```
practic.js > add
1 let num1 = 12;
2 let num2 = 11;
3 function add(a, b){
4   return a+b;
5 }
6 console.log("addition =", add(num1, num2));
7 console.log("radhika");

PS C:\Users\itlab\Desktop\radha tyit\mern stact> node pract1d.js
addition = 23
radhika
```

e. Demonstrate use of operators and control statements in JavaScript

```
1 let a = 12;
2 let b = 11;
3 console.log("\NCOMPARISION OPERATORS");
4 console.log("a > b:", a > b);
5 console.log("a < b:", a < b);
6 if (a > b){
7     console.log("a is greater then b");
8 }
9 else {
10     console.log("b is greater then a");
11 }
12 let day = 2;
13 switch(day){
14     case 1:
15         console.log("monday");
16         break;
17     case 2:
18         console.log("tuesday");
19         break;
20     case 3:
21         onsole.log("wednesday");
22         break;
23     default:
24         console.log("invalid day");
25 }
26 console.log("radhika");|
```

```
PS C:\Users\itlab\Desktop\radha tyit\mern stact> node practice.js
\NCOMPARISION OPERATORS
a > b: true
a < b: false
a is greater then b
tuesday
radhika
```

f. Write programs using loops and template literals.

```
1 let num = 5;
2 console.log(`Multiplication table of ${num}`);
3 for (let i=1; i<=10; i++){
4     console.log(`${num} x ${i} = ${num*i}`);
5 }
```

```
PS C:\Users\itlab\Desktop\radha tyit\mern stact> node pract1f.js
Multiplication table of 5
5 x 1 = 5
5 x 1 = 10
5 x 1 = 15
5 x 1 = 20
5 x 1 = 25
5 x 1 = 30
5 x 1 = 35
5 x 1 = 40
5 x 1 = 45
5 x 1 = 50
```